Tunisian Republic
Ministry of Higher Education
and Scientific Research
University of Carthage
Higher Institute of Technologies
of Information and Communication

LICENCE IN COMPUTER SYSTEMS ENGINEERING

SPECIALTY: NETWORK AND SYSTEMS ENGINEERING

Construction of a Cisco IP Fabric for the IT Data Center Network based on the BGP EVPN-VXLAN Standard

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Prepared by

Chams Edin Fares
Wajih Ben Zekri

Carried out at Tunisie Telecom

Publicly defended on [insert date here], 2026 before the jury composed of:

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Academic Year: 2025–2026

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Acknowledgments

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Dedications

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List of Abbreviations

TT	Tunisie Telecom
BGP	Border Gateway Protocol
EVPN	Ethernet Virtual Private Network
VXLAN	Virtual Extensible Local Area Network
VNI	VXLAN Network Identifier
VTEP	VXLAN Tunnel Endpoint
MP-BGP	Multiprotocol BGP
CLOS	Clos topology
OSPF	Open Shortest Path First
AI	Artificial Intelligence
DC	data center
IS-IS	Intermediate System to Intermediate System
DCI	Data Center Interconnect
TT	Tunisie Telecom
AS	Autonomous System
ASN	Autonomous System Number
VM	Virtual Machine
SDN	Software Defined Network
IBN	Intent-Based Networking
CI/CD	Continuous Integration/Continuous Delivery
AIOps	Artificial Intelligence for IT Operations
CLI	Command Line Interfaces

General Introduction

The exponential rise of cloud computing, hyperscale operations, and AI has completely revolutionized what infrastructure looks like today. AI platform, from distributed training to real-time inference, drive network requirements that are an order of magnitude beyond traditional enterprise applications. Data centers have moved from simple storage repositories to becoming the engines of the digital economy, where network scalability is now the determining factor of infrastructure success and failure.

The traditional three-tier architecture is rapidly becoming obsolete, unable to effectively address East-West traffic and mobility requirements of AI-based applications and services. Inefficiencies such as Spanning Tree's bandwidth waste, unstable Layer 2 domains, and site boundaries are becoming major bottlenecks for virtual machine mobility and multi-tenancy. For a national operator such as Tunisie Telecom, these are not just conceptual limitations, but real-world inhibitors of network scalability and reliability in critical telecommunications infrastructure.

This project is designed to deploy a BGP EVPN-VXLAN IP Fabric across various data centers using Cisco Nexus 9000 switches and the Nexus Dashboard Fabric Controller (NDFC) product. This new approach to the legacy architecture is designed to overcome the limitations of VM mobility, Layer 2 extension, and tenancy for Tunisie Telecom. In the end, it provides a fabric that is suitable for future AI-driven applications without the need for further fundamental changes.

This report is assembled into five chapters:

∗ **Chapter 1:** Introduces Tunisie Telecom as the host organization, establishes the technical and operational context of the project.

∗ **Chapter 2:** Builds the theoretical foundation of the project.

∗ **Chapter 3:** Examines the evolution of network automation, establishes the case for infrastructure programmability at scale.

∗ **Chapter 4:** Presents the complete multi-site IP Fabric architecture.

∗ **Chapter 5:** Covers the full deployment of the solution within an EVE-NG emulation environment.

The work presented here is not an academic exercise but a concrete response to the infrastructure challenges facing one of Tunisia's most critical telecommunications operators.

Chapter 1

Presentation of the Project Framework

1.1 Introduction

This opening chapter establishes the foundational context for the project. It begins by introducing the host organization, Tunisie Telecom, detailing its historical evolution, organizational mission, and its current standing in the telecommunications market. Furthermore, this chapter outlines the specific project environment, the technical challenges faced by the operator, and the strategic objectives that the proposed Data Center Fabric aims to achieve.

1.2 Presentation of the Company

1.2.1 Tunisie Telecom

Tunisie Telecom [1] is a leading and historic telecommunication company in the country. It is a public limited company with a large customer base of over six million subscribers in the fixed network, mobile phone, and high-speed internet services, both locally and globally. The company is currently operating as a joint venture with the Tunisian state holding a 65% majority stake and Dubai Holding [2] holding the remaining 35%.

Figure 1.1: Headquarters of Tunisie Telecom [1]

1.2.2 History of Tunisie Telecom

TT[1] was officially formed on April 17, 1995, as a strategic move of the Tunisian state to bring its communication services under a public limited company. This period has seen a move to digital exchanges as well as a huge expansion of the fixed-line network.

1.2.3 Corporate Missions & Organizational Chart of the Company

1. **Corporate Missions**
   TT[1] operates under a mandate to secure and advance the national digital landscape. Its core missions include:

   - Network Deployment: Developing and installing advanced telephony and data transmission networks.
   - Security & Reliability: Ensuring the homogeneity, confidentiality, and high availability of information systems nationwide.
   - Data Management: Implementing and managing complex databases for a customer base exceeding 6 million subscribers.
   - Consultancy: Providing expert telecommunications advice to public and private entities, both nationally and internationally.

2. **Organizational Chart**
   The organizational structure of Tunisie Telecom is designed to support its mission as a national technological leader while ensuring operational agility

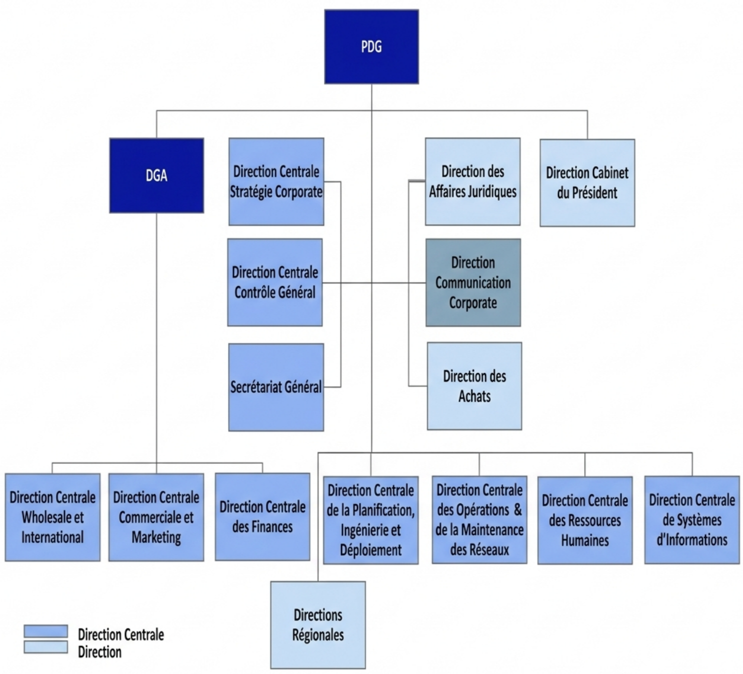

Figure 1.2: Organizational Chart

- CEO (PDG): Ensures the overall strategic direction of the company, defines the long-term vision, and oversees the execution of national telecommunications policies
- Deputy CEO (DGA): Assists the CEO in daily management and acts as a substitute in operational coordination functions.
- Office of the President: Manages the strategic agenda and institutional relations for senior management.

- Corporate Strategy Central Directorate: Responsible for strategic planning, competitive intelligence, and the evolution of the operator's business model.
- General Audit Central Directorate: Guarantees internal auditing, process transparency, and risk management.
- General Secretariat: Ensures administrative coordination and monitors the decisions made by the Board of Directors.
- Legal Affairs Directorate: Ensures compliance with the telecommunications regulatory framework and manages litigation.
- Corporate Communication Directorate: Drives brand image and institutional communication with the public and partners.
- Procurement Directorate: Centralizes supply management and relations with technology providers.
- Wholesale and International Central Directorate: Manages international transit infrastructure (such as submarine cables) and relations with foreign operators.
- Commercial and Marketing Central Directorate: Develops offerings for residential and business segments and defines pricing policies.
- Finance Central Directorate: Ensures budget management, accounting, and the financial viability of technological investments.
- Planning, Engineering, and Deployment Central Directorate: Responsible for network architecture design and national coverage expansion.
- Network Operations & Maintenance Central Directorate: Guarantees 24/7 infrastructure availability and the resolution of technical incidents.
- Human Resources Central Directorate: Manages skills development and training for the employees.
- Information Systems Central Directorate (ISD / DSI): it manages the IT infrastructure, servers, and data security for the subscribers.
- Regional Directorates: Provide a local presence for Tunisie Telecom across the country's 24 governorates, ensuring quality of service at the local level.

1.2.4 Partners of Tunisie Telecom

To maintain its technological edge, TT[1] collaborates with a diverse group of global and local leaders:

Figure 1.3: Partners

- **Dubai Holding:** Strategic international investor holding a 35% stake in the company.
- **Huawei Marine:** Technical partner for international submarine fiber-optic infrastructure

- **The Tunisian Post:** Collaborative partner for national mobile financial and payment services.
- **GO Malta:** Strategic subsidiary for European market integration
- **Cisco & Microsoft:** Key technology providers for Data Center modernization and cloud integration

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1.2.5 Solutions and Services Offered by Tunisie Telecom

TT[1] stands as the primary telecommunications service provider in Tunisia. It offers a comprehensive range of telephony, Internet, and television services to its diverse clientele.

Service Category	**Description**
Fixed Telephony	Provision of landline services to both residential and professional customers across Tunisia.
Mobile Telephony	Mobile services under the "Tunisie Telecom Mobile" brand, encompassing voice calls, SMS, and data services.
Internet Services	High-speed broadband connectivity via ADSL, Wi-Fi, and fiber optics, as well as Voice over IP (VoIP) solutions.
Television Services	A diverse suite of television offerings, including IPTV, Direct-to-Home (DTH), Video on Demand (VOD), and online streaming.
Enterprise Solutions	Specialized telecommunications for businesses, featuring call center services, private network management, video conferencing, and high-bandwidth connectivity.
Financial Services	Financial offerings such as money transfers and mobile or electronic payment systems.

Table 1.1: Portfolio of Solutions and Services provided by Tunisie Telecom

1.3 Presentation of the Project

As Tunisie Telecom continues to expand its digital infrastructure to meet growing national demand, the underlying network architecture of its data centers has become a critical point of re-evaluation. This section presents the technical context that motivated this project, beginning with a broad examination of modern data center networking challenges before narrowing to the specific constraints faced by the operator.

1.3.1 Data Center Networking Context and Limitations

A data center is essentially built on three pillars of compute, storage, and networking, with networking being the key component that determines the speed of communication and prevents bottlenecks. The importance of networking has become increasingly critical in recent times, especially owing to factors like server virtualization, cloud-native applications, and especially artificial intelligence, which has caused a shift from traditional north-south communication patterns to east-west communication patterns involving huge volumes of data between GPU clusters and storage systems. The traditional three-tier data center architecture, which is commonly used by various data centers around the world, including Tunisie Telecom, is found wanting in this regard.

The fact that it utilize Spanning Tree Protocol (STP) and blocks redundant paths by disabling them is causing huge bandwidth wastage, and its hierarchical structure is causing scalability and oversubscription issues. These issues become more complex when data centers are distributed in various geographical locations, as this is not a feature that this data center architecture was designed for, and it is a necessity for data center environments. All of this has caused Tunisie Telecom to rethink its data center networking infrastructure.

1.3.2 Critique of the Existing System

Tunisie Telecom currently operates its data center network infrastructure based on the classical three-tier architecture described above. Given the operator's scale, its national reach, and the growing complexity of the workloads it must support, the limitations inherent to this model translate into concrete operational and business constraints that can no longer be deferred. The following points shows the most important issues:

- **STP-induced bandwidth waste:** Redundant physical links between sites remain blocked during normal operation, resulting in underutilized capacity precisely where throughput is most needed.
- **VM mobility constraints:** Virtual machine migration between geographically separated sites is either severely limited or operationally impractical, directly impacting workload balancing and business continuity capabilities.
- **L2 extension instability:** Stretching Layer 2 domains across multiple sites introduces broadcast instability, MAC table overflow risks, and expanded failure domains that are difficult to contain at the operator's scale.
- **Inter-site latency:** The multi-hop nature of the three-tier model adds forwarding delay between sites, making the infrastructure increasingly inadequate for latency-sensitive and AI-driven distributed workloads a dimension that is central to Tunisie Telecom's ongoing digital transformation strategy.
- **VLAN Scalability Limit:** The 4,096 VLAN limit is insufficient for modern multi-tenant environments, failing to provide the granular traffic segregation required by a diverse client base ranging from large enterprises to public institutions that are accessing Tunisie Telecom's infrastructure.
- **Manual configuration burden:** The lack of a centralized automation system forces network engineers to manually configure every device in the network, which can lead to configuration drift and slow response to changing network infrastructure.

These constraints create a compelling argument that a radical rethink of Tunisie Telecom's data center network design is warranted. The proposed solution addresses each of these constraints directly.

1.3.3 Proposed Solution

Where traditional architectures have reached their limits, modern networking standards offer a fundamentally different approach. The solution proposed in this project is the design and deployment of a BGP EVPN-VXLAN IP Fabric spanning multiple sites — an architecture that does not merely patch the weaknesses of the existing model but replaces its core assumptions entirely. It is based on four tightly integrated building blocks:

- BGP EVPN-VXLAN IP Fabric: VXLAN allows the logical network to be decoupled from the physical infrastructure by encapsulating the L2 frames within the UDP packets, thereby removing the geographic extension of the L2 networks and providing seamless VM mobility. BGP EVPN preform as the control plane, distributing MAC and IP reachability proactively across all fabric nodes — suppressing broadcast traffic, scaling to tens of thousands of endpoints, and delivering the multi-tenancy granularity that Tunisie Telecom's diverse client base demands through a 24-bit VNI space of over 16 million logical segments.

- Spine-Leaf Topology: Replacing the three-tier hierarchy with a flat two-hop architecture where every Leaf connects to every Spine. This also supports Equal Cost Multi-Path forwarding on all links simultaneously, eliminates bandwidth waste caused by STP, and provides predictable low latency — a non-negotiable requirement for AI-driven distributed workloads at the heart of Tunisie Telecom's infrastructure vision.
- Cisco Nexus 9000: The hardware underpinning of the fabric, providing native line-rate BGP EVPN VXLAN forwarding in silicon, ensuring that encapsulation and processing do not introduce latency or performance degradation of a kind that would undermine our latency objectives.
- Cisco NDFC (Nexus Dashboard Fabric Controller): The centralized automation and orchestration solution for the entire underlay and overlay fabric from a single management plane, directly eliminating manual configuration effort, configuration drift, and policy enforcement inconsistencies.

The implementation of this solution is based on a structured and iterative methodology, as described in the methodology section below.

1.4 Conclusion

This chapter sets the scene for the project, starting off with a brief overview of the company, Tunisie Telecom, and their service provision to over six million subscribers, followed by a discussion of the technological reasons for architectural change. Whilst the three-tier architecture has served its purpose in the past, it is now seriously flawed in its ability to meet current needs such as geographical distribution, VM mobility, L2 extension, and scalability in a world of artificial intelligence. This project goals are to overcome these limitations and negate the complexity of configuration by implementing a BGP EVPN-VXLAN IP Fabric solution using Cisco Nexus 9000 switches and NDFC, simulating the environment in EVE-NG. The theoretical basis for this architecture is described in the following chapter.

Chapter 2

Foundations and Protocols

2.1 Introduction

The chapter starts with a description of the evolution of data centers from simple server rooms to the backbone of the digital economy, with a focus on how network architectures have evolved to accommodate the growing demand for these solutions. It also introduces the theoretical framework for network layering and operational planes, which are necessary for understanding the protocols used in this project. The chapter then delves into the details of the Spine-Leaf, IP Fabric, and the important abstraction of underlays and overlays. The key protocols used in this project, which are also the core protocols used in this chapter, are VXLAN for the data plane, BGP for the underlay, and BGP EVPN for the overlay control plane. Finally, the chapter includes a comparative analysis of other technologies used in the field, which validates the use of the protocols in this project.

2.2 The Data Center

2.2.1 Evolution, Definition and Role

The idea of a data center can be traced back to the early 1960s when organizations started centralizing their computer resources in a specific room where mainframe computer systems were located[3]. This is origin of the idea of a data center came about. A data center is a centralized computer environment that is designed to house computer systems that are usually expensive and sensitive. In this era, a data center is considered a single computer that is used by an entire organization. This is because users access this computer using terminals and not individual computers.

The development of the client/server model in the 1980s changed everything[4]. In this era, every computer had its own role. A server is used for specific purposes such as storing files, printing documents, and running database management systems. A client is used for interacting with users. In this era, data centers became larger in size because organizations were using multiple servers to support their computer systems. This is when the need for structured cabling became apparent.

The explosion of Internet services in the 1990s accompanied in a new era for DCs in terms of scale and criticality[5]. Organizations started to host Internet-facing services, which required availability. The idea of data centers as business-critical facilities, as opposed to back-office operations, really took hold in this time.

The 2000s saw the advent of server virtualization, starting with VMware's ESX Server release in 2001[6]. A physical server could now host multiple virtual machines, greatly improving hardware utilization and reducing physical data center space. Networking challenges also started to manifest themselves in this time, as virtual machines needed to talk across physical network partitions. The limitations of network architectures were first felt in this time. Cloud computing was the dominant technology paradigm of the 2010s, and cloud providers such as Amazon Web Services, Microsoft Azure, and Google Cloud Platform revolutionized the concept of a data center[7]. Infrastructure was offered as a service, where it was used on demand, metered by consumption, and located in a facility of unprecedented scale. Hyperscale data centers with hundreds of thousands of servers became a reality, and the networking infrastructure that interconnected them followed suit.

Today, the data center is at the center of a new inflection point driven by the advent of Artificial Intelligence[8]. AI training workloads require massive parallelism, high-bandwidth interconnects between GPU clusters, and ultra-low latency communication between distributed compute nodes. The modern DC is formerly just a passive repository for the world's increasingly sophisticated hardware; it is a programmable infrastructure that provides compute, storage, and networking as unified software resources to applications that operate at a scale that is global in scope.

Figure 2.1: Computing Evolution Era Timeline

DC has been formally defined as a special facility that centralizes the information technology infrastructure of an organization in a highly available and controlled environment[9]. Beyond the hosting of hardware devices, the contemporary data center has become the essential operational platform for the information age, supporting every service, transaction, and communication in the interconnected world.

2.2.2 Tier Classification and Foundational Pillars

The Uptime Institute established the data center Tier Classification Standard to provide a consistent framework for evaluating infrastructure reliability and redundancy[10]. The standard defines four tiers, each representing a progressively higher level of fault tolerance and availability, as shown in the table All DCs, regardless of their tier level, are constructed around three basic pillars,

Tier	Availability	Annual Downtime	Redundancy	Key Characteristics
Tier I	99.671%	28.8 hours	None	Single path for power/cooling, no redundant components.
Tier II	99.741%	22.0 hours	Partial	Redundant capacity components, single distribution path.
Tier III	99.982%	1.6 hours	N+1	Multiple paths (one active), concurrently maintainable.
Tier IV	99.995%	26.3 minutes	2N	Fully redundant, fault tolerant, simultaneous maintainability.

Table 2.1: Uptime Institute Data Center Tier Classification[10]

which interact with each other and define the overall performance of the DC[11].

- Compute refers to the processing layer of the data center, which is the set of servers, processors, and increasingly, the set of graphics processing units that perform the workload the DC was created to perform. Modern compute infrastructure is almost universally virtualized, which means that the underlying physical infrastructure is divided into several isolated environments, which can exist independently of the underlying infrastructure.

- Storage refers to the persistent memory layer of the DC, which is the set of storage devices that retains data across compute cycles, failures, and migrations. Storage architectures vary from direct storage attached to individual servers, through shared storage solutions, and up to the object storage solutions that span entire data center fabrics. The performance of the storage solution defines the rate of data that can be read and written, which makes storage a critical factor in determining the overall service quality.

- Networking is the interconnect layer that binds together compute and storage into a coherent operational environment[12]. It impacts the speed at which a compute node can access a storage system, the efficiency of intercommunication between two virtual machines, and the reliability of intercommunication between the data center and the external world. Of the three pillars, networking is the most architectural in nature; decisions made here cascade across the entire DC and constrain the effectiveness of compute and storage. It is this pillar of the DC that our project impacts.

2.3 From Three-Tier to IP Fabric

The architectures used to network these DCs have also undergone a revolutionary change over the past two decades or so. What started as a simple hierarchical model has now evolved into the programmable and distributed fabric model due to the influence of software-defined networking and virtualization of network services and hyperscale DCs.

2.3.1 The Three-Tier Model and Its Limits

For several decades, the three-tier architecture was the de facto standard for network architecture in the DC and enterprises alike[13]. The three-tier architecture was based on a three-tier model consisting of Access, Distribution, and Core, which provided a structured approach for interconnecting devices to the network, with each tier having a well-defined role in the forwarding model. The Access layer is the essential connection of end devices such as servers, workstations,

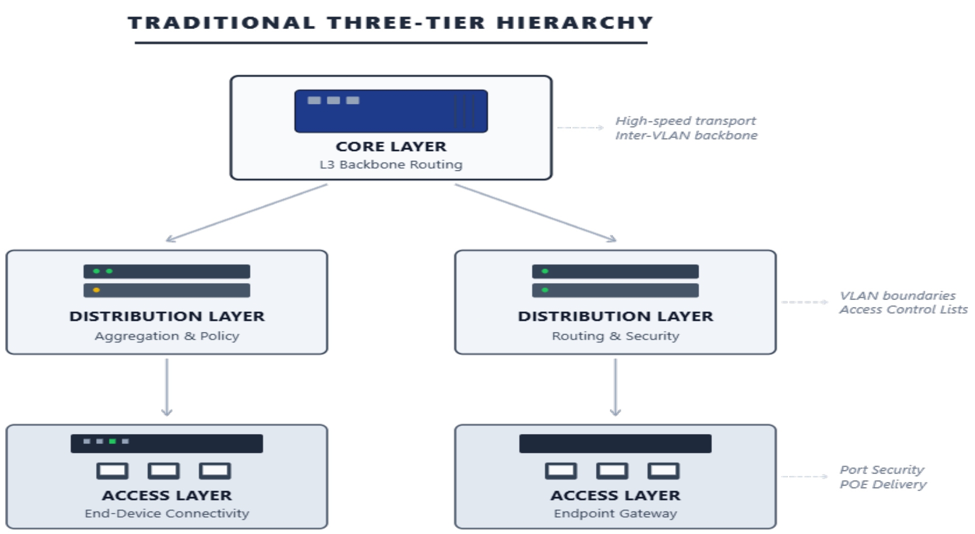

Figure 2.2: Three-Tier Architecture Model

and storage devices to the network fabric. The Distribution layer is responsible of aggregating Access layer traffic and routing policy implementation. Lastly, the Core layer is in charge of high-speed connectivity between Distribution blocks and also deals with site or internet-bound traffic. The three-tier network model is highly popular and widely applied, although it has some structural limitations, especially as the needs of the DCs increase. The pain points of the three-tier network model include:

- STP Dependency: Spanning Tree Protocol is designed to block redundant links to prevent loops. This wastes active bandwidth and causes inefficiencies throughout the entire Access and Distribution hierarchy.
- Oversubscription: The aggregating nature of the network flows upward through the hierarchy creates potential bottlenecks in the Distribution and Core layers.
- East-West Traffic Problem: The original design of the network model is to handle north-south flows of clients and servers. However, the virtualized world creates server-to-server flows, and the inefficiencies in the forwarding tables have exposed fundamental flaws.
- Scalability Ceiling: To increase the network, more hierarchy is added, which increases the complexity.
- Large Failure Domains: A failure in the Distribution or Core layer can impact the entire segment of the network.

2.3.2 SDN: Birth and Core Concepts

The limitation of the three-tier model has led the networking industry to adopt a completely new paradigm, which separated the intelligence of the network from its forwarding hardware. Software-Defined Networking is a new paradigm in the field of networking, which originated as a research project in 2008 by Martin Casado and Scott Shenker of Stanford University and introduced the concept of OpenFlow as a protocol for the centralized control of network forwarding behavior.

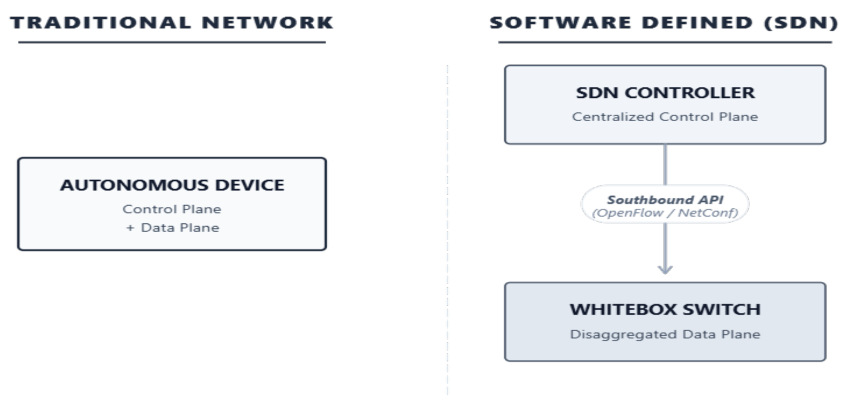

Figure 2.3: Software-Defined Networking Architecture

The SDN architecture has three distinct layers[14]. The Application Layer resides at the top and communicates with the controller through Northbound APIs. The Northbound APIs are typically REST APIs. The Application Layer communicates its intent to the controller. The Application Layer does not interact with the network devices. The second layer of the SDN architecture is the Control Layer. The SDN controller accommodated in this layer. The SDN controller has a global perspective of the network topology. The third layer of the SDN architecture is the Infrastructure Layer. The network devices reside in this layer. The network devices which are forwarding devices. The network devices are programmed by the SDN controller. The network devices are programmed through Southbound APIs. The Southbound APIs are typically OpenFlow. This decoupling of control and data planes was a conceptual leap forward, as for the first time network behavior was programmable from a central location rather than being configured on a per-device basis [15]. It also introduced the idea of dynamic application-aware traffic management at a scale and speed that traditional distributed protocols couldn't deliver. next section

2.3.3 Network Functions Virtualization

Coexisting alongside SDN, NFV was first formally defined by a group of major telecommunications operators through a white paper submitted to the ETSI standards body in 2012[16]. While SDN was addressing the programmability of the data plane, NFV was addressing a different issue related to the dependency of network functions on proprietary and dedicated hardware. NFV was based on the idea of running firewalls, load balancers, intrusion detection systems, and WAN optimizers not on proprietary and dedicated hardware but on standard commercial-off-the-shelf servers. In other words, NFV was based on the idea of running network functions as software on top of servers. This would enable network functions to be deployed, scaled, and moved around dynamically, similar to how virtual machines are deployed, scaled, and moved around on top of physical servers. NFV and SDN are not competing technologies but are complementary to each other.

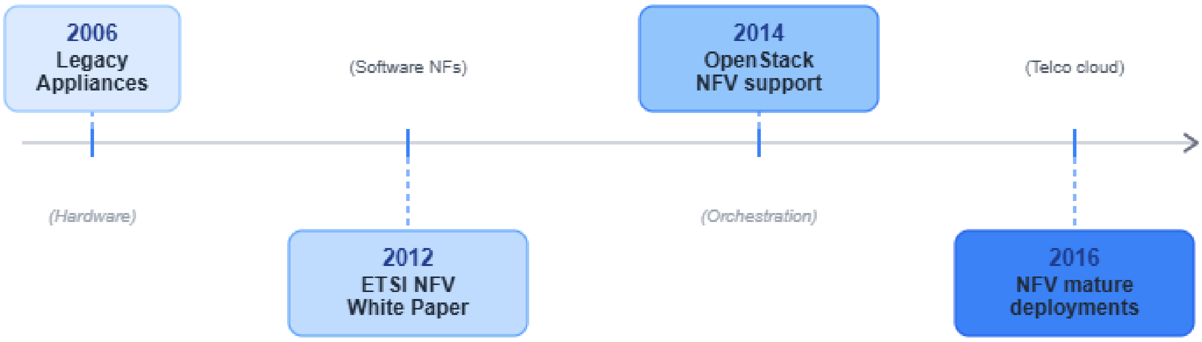

Figure 2.4: Network Functions Virtualization Timeline

2.3.4 The Emergence of the IP Fabric

While SDN and NFV have shown much promise, the actual implementation of OpenFlow-based networks showed severe limitations in a production network environment [17].In such a network, the controller is the only point of failure as the network was expanding in scale. The entire network endangered if the controller was to fail. The global network view was operationally complex with thousands of devices. The concept of Proprietary SDNs was introduced by various vendors like Cisco's ACI and VMware's NSX. They introduced a controller-based SDN solution that provided abstraction of the network with distributed forwarding in the devices [18]. These implementations suggested the path toward the hybrid model, which was to be the hallmark of the data center networking paradigm in the future, i.e., one that offered the advantages of distributed protocol intelligence in the forwarding plane, along with the advantages offered by the central visibility and automation offered in the management plane. The IP Fabric model was the result of this development, where the flat Spine-Leaf physical topology was integrated with standards-based overlay protocols along with the central management plane [19]. The IP Fabric model does not replace the distributed protocol model but instead leverages the advantages offered by the use of BGP for the distributed control plane along with the flexibility offered by the use of VXLAN for the overlay, thus achieving the programmability offered by SDN without the risk of single-point failure. The IP Fabric is the final form of the entire evolution path, from the strict hierarchy of the three-tier model through the centralized intelligence of the SDN to the design that is both distributed, scalable, and programmable[20]. This is the abstraction which upon the solution proposed in the project is based.

2.4 Network Layering and Operational Planes

Prior to discussing the protocols and topologies that form the basis of this project, it is necessary to first identify the conceptual framework through which network communication is defined and implemented. Two primary models form the basis of the framework, namely the OSI reference model and the TCP/IP model, from which the three operational planes are derived.

2.4.1 Layering Models

The OSI model, as proposed by ISO in 1984, divides the communication process into seven abstraction layers[21]. Though it remains the normative reference for understanding the behavior of communication protocols and troubleshooting, it is not commonly implemented. The TCP/IP model, based on the ARPANET development and IETF standards, proposes a four-layer model that is actually implemented by real protocols[22] In DC engineering, the TCP/IP model is the reference

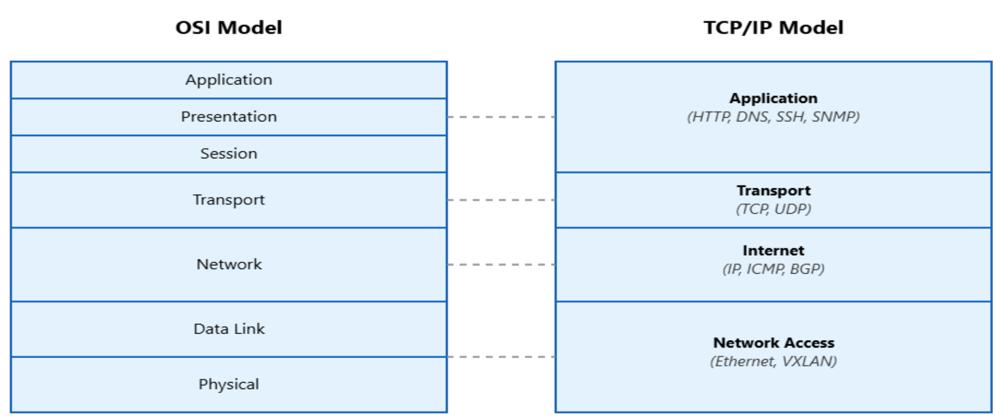

Figure 2.5: OSI vs TCP/IP Layering Models

for operations, with every protocol used in the data center fabric mapped to one of four layers[23]. The OSI model is used as an analytical tool, particularly when thinking about where encapsulation boundaries occur or how an overlay protocol interacts with the underlying physical network.

2.4.2 The Three Planes and Their Role

Aside from protocol layering, today's network infrastructure can be conceptually divided into three planes, each with a specific category of network function[24].

Plane	TCP/IP Layer	Role	Technology
Data	Network Access + Internet	Packet forwarding, encapsulation, switching at line rate	VXLAN
Control	Internet + Transport	Topology computation, routing protocol signaling, reachability distribution	BGP EVPN
Mgmt	Application	Configuration, monitoring, policy orchestration, automation	Cisco NDFC

Table 2.2: Operational Planes Mapped to This Project

- **Data Plane**: This is the plane where the actual movement of packets happens. This is where the network operates, typically at the hardware level, through ASICs that make forwarding decisions without any involvement of software[25]. In this project, VXLAN is used as a data plane, where it encapsulates the original Layer 2 packets into a UDP packet, allowing it to traverse any IP network.

- **Control Plane**: This is the plane at which the network topologies are built and maintained. This is where routing protocols operate, where optimal paths are determined, and where network reachability information is disseminated to all the network elements that participated in the network [26]. In this project, BGP EVPN was used as a control plane where network MAC and IP information were disseminated to all VTEP elements in the network without using flood and learn.

- **Management Plane**: This layer is situated at a higher level than the other two and is the point of interface between the network and the network operator. This is where network operators configure, monitor, and automate the network [27]. The Cisco NDFC is situated at this plane in this project.

This division of the three planes is not merely theoretical but is the fundamental principle that allows data center networks to be at the same time scalable, programmable, and fault-tolerant. The data plane and the in-flight data do not get interrupted in the event of a failure in the control plane, and similarly, the management plane operations do not impact the data plane or the routing operations.

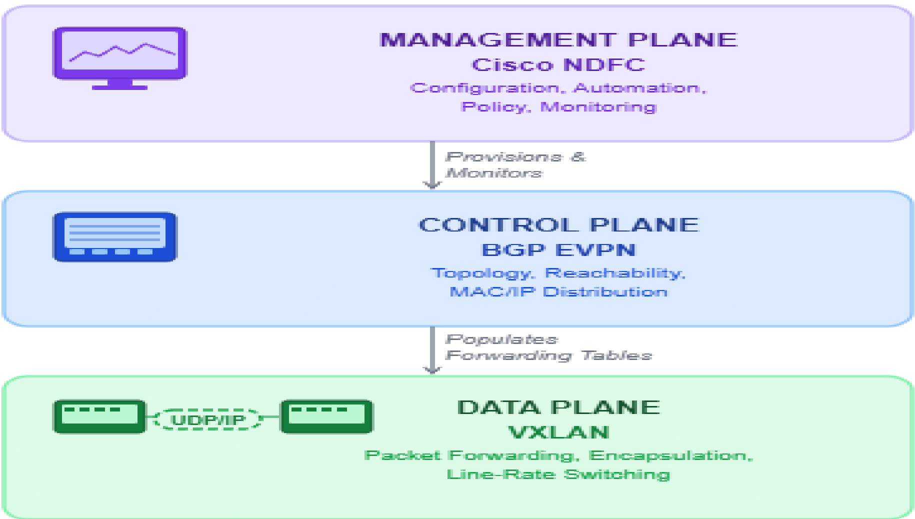

Figure 2.6: Three Planes Interaction Diagram

2.5 Spine-Leaf and IP Fabric

The Spine-Leaf topology is the fundamental basis upon which IP Fabrics are designed. The adoption of the Spine-Leaf topology in data center networking signifies a significant shift away from the traditional hierarchical structure of the three-tier topology, with its emphasis on vertical aggregation, to a flat, symmetric, and fully meshed topology, which is optimized for performance at scale.

2.5.1 Origins, Roles and Design Principles

The Spine-Leaf topology is based on a variant of a Clos topology, a multistage switching topology proposed by Charles Clos at Bell Labs in 1952 to optimize telephone circuit switching via a non-blocking interconnection topology[28]. Re-discovered by hyperscalers such as Google and Facebook in the late 2000s as a solution to the exploding east-west traffic problem created by distributed computing workloads, the Clos topology has evolved into the two-tier Spine-Leaf topology now ubiquitous in data center networks[29]. The architecture is divided into two tiers

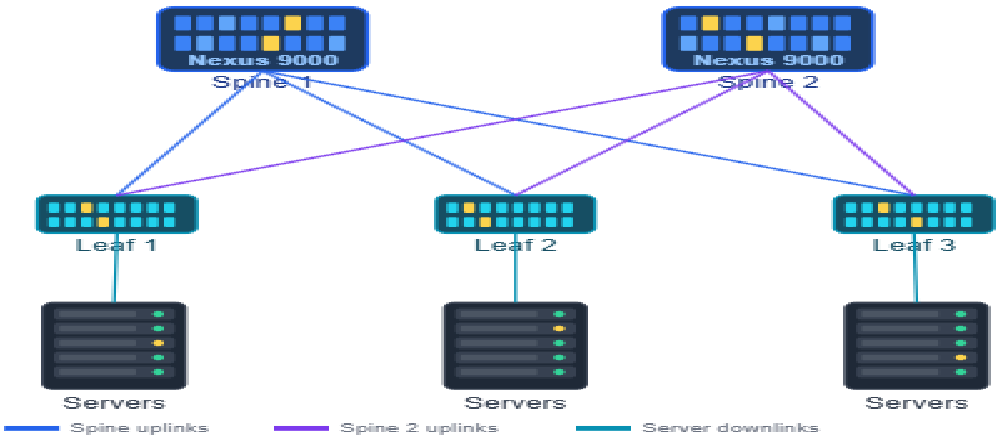

Figure 2.7: Spine-Leaf Topology

with distinct responsibilities[30]. The first tier consists of spine nodes, which form the backbones of the fabric. They are responsible for routing between Leaf nodes. They are exclusively responsible for routing. They never connect to an end device. The second tier consists of Leaf nodes. They are access points for the fabric. A server, storage device, firewall, or gateway is only connected to a Leaf node. Importantly, no direct connections are allowed between Leaf nodes or between Spine nodes. Any two nodes can only communicate through exactly two hops: the source Leaf, the destination Leaf, and one spine node in between[30]. This ensures that latency is uniform throughout the fabric, irrespective of the communicating devices.

2.5.2 Oversubscription and Non-Blocking Ratios

Oversubscription is defined as the ratio of device bandwidth to uplink capacity[31]. In traditional three-tier networks, this oversubscription occurs in Access, Distribution, and Core layers.
The Spine-Leaf network design avoids this issue by allocating Leaf to Spine bandwidth in direct proportion to server access capacity[32]. This provides a non-blocking network where uplink capacity matches access demand, so application performance is not constrained by network architecture but rather by actual application demand.

2.5.3 Equal-Cost Multi-Path

ECMP helps the full mesh Spine-Leaf topology function by using all paths at once[33]. This is different from the Spanning Tree Protocol, where it de-activates all paths to avoid loops in the network[34].
Since all paths between the Leafs and the Spine are now active paths, ECMP hashing of the flow based on flow identifiers such as IP and ports allows automatic load balancing and flow consistency[35]. This allows the network to scale linearly with the number of paths provided.

2.5.4 IP Fabric and Pod Architecture

IP Fabric is the logical structure resulting from the combination of the physical Spine-Leaf network and a pure Layer 3 network infrastructure, in which all paths between the switches are routed point-to-point interfaces, all Layer 2 domains are confined to the server-facing ports of the Leafs, and all communication between Leafs is routed using IP[19]. This completely removes broadcast domains from the core of the fabric and removes the instability and scalability issues associated with them, replacing them with the ability to run scalable routing protocols such as BGP. As fabrics expand beyond a single set of Spine and Leaf nodes, the Pod provides a scaling unit as defined in[36]. The Pod is a self-contained Spine-Leafe fabric consisting of a fixed number of Spine and Leaf

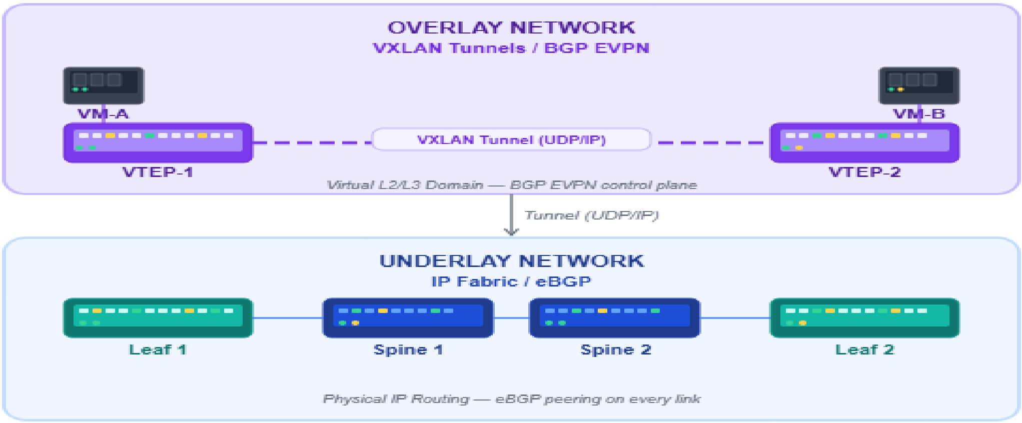

Figure 2.8: IP Fabric Underlay Concept

nodes as an independent forwarding domain. Multiple Pods connect to one another through a Super Spine layer, enabling horizontal scaling of the fabric through the addition of multiple Pods rather than increasing switching capacity. This is particularly relevant to the multi-site nature of this project's design intent—each data center site could be thought of as an independent Pod, connected via a DCI layer extending the IP Fabric beyond physical site boundaries[37].

2.6 Underlay and Overlay

The IP Fabric architecture is divided into two layers: underlay and overlay. The relationship between these two layers is essential in understanding how VXLAN and BGP EVPN operate in tandem as an integral part of the solution proposed in this project.

2.6.1 The Underlay/The Overlay Network

The underlay network is defined as the physical IP infrastructure that provides basic IP reachability between all nodes in the IP fabric network[38]. This network includes Spine and Leaf switches connected via routed point-to-point links running a routing protocol, in this case, eBGP, to exchange IP reachability information among all VTEPs in the network. There is no knowledge in this network of tenants, virtual machines, or logical network segments; its sole aim is to ensure that any node in the network is reachable via a routed IP path from any other node in the network. The overlay network is a logical network running on top of the underlay network. It provides virtual network connectivity between network endpoints irrespective of their physical position within the network fabric[39]. It carries traffic within a tunnel, where the traffic is sent through the underlay network as normal IP packets. The underlay network is completely unaware of the inner packets. In this project, the VXLAN protocol is used for the encapsulation process, and the control plane is provided by the BGP EVPN.

2.6.2 Why the Separation Matters

The decoupling of the overlay from the underlay is the architectural choice that solves the fundamental issues identified in Chapter 1, as it allows the logical network topology to grow and expand independently of the physical topology beneath it[37]. A virtual machine move is possible, a new tenant segment is created, or a physical link fails, and none of these requires changes to the other topology. The underlay is the steady and constant IP transport, and the overlay is where all the complexity is handled. This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control

Characteristic	**Underlay**	**Overlay**
Layer	*L3* — Pure IP routing	*L2/L3* — Virtual logical network
Protocol	eBGP	VXLAN + BGP EVPN
Awareness	Physical topology only	Tenant traffic and endpoints
Responsibility	IP reachability between VTEPs	Virtual connectivity between VMs
Failure Impact	Affects all tunnels	Affects specific tenant segments
Scalability Unit	Physical links and nodes	VNI and VRF count

Table 2.3: Underlay vs Overlay Characteristics

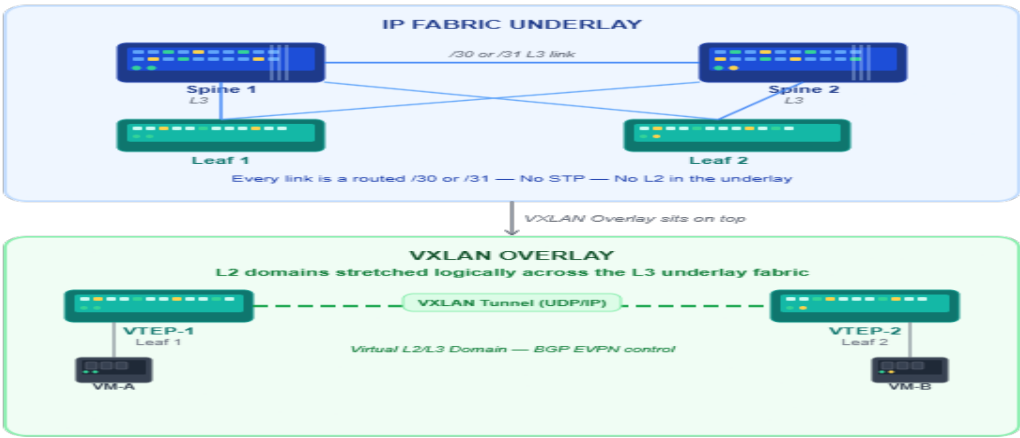

Figure 2.9: Underlay vs Overlay Diagram

plane of the solution presented in this project[40].

2.7 VXLAN

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.7.1 Context, Motivation and History

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.7.2 Encapsulation, VNI and Packet Format

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.7.3 The VTEP

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.7.4 Flood and Learn, Multicast and BUM Traffic

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.7.5 Lexicon

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.8 BGP

2.8.1 History, Evolution and DC Adoption

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.8.2 Core Mechanics

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.8.3 iBGP vs eBGP

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.8.4 Path Selection

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.8.5 Address Families

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.8.6 Route Reflectors

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.8.7 Confederations

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.8.8 Lexicon

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.9 BGP - EVPN

2.9.1 Context, History and Definition

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.9.2 From VPLS to EVPN

2.9.3 EVPN Route Types

This is exactly what VXLAN does to extend a Layer 2 domain over any IP network despite the distance between the sites, and what BGP EVPN does to transport MAC and IP reachability without the need to flood the network, creating the complete data and control plane of the solution presented in this project[40].

2.9.4 How EVPN Drives VXLAN

2.9.5 ARP Suppression

2.9.6 MAC Mobility

2.9.7 L2VNI and L3VNI

2.9.8 Symmetric vs Asymmetric IRB

2.9.9 Anycast Gateway

2.9.10 Multi-Tenancy and VRF Design

2.9.11 Lexicon

2.10 Comparative Study and Technology Selection

2.10.1 Selection Methodology

2.10.2 Underlay Alternatives

2.10.3 Overlay Alternatives

2.10.4 Control Plane Alternatives

2.10.5 Platform Comparison

2.10.6 Retained — Cisco Nexus 9000

2.11 Conclusion

Chapter 3

Network Automation and Emulation Platforms

3.1 Introduction

This chapter will discuss the ecosystem in question from two different perspectives. One is network automation, where the progression from manual CLI-based operations towards INT-based networks is discussed, the architectural foundations of network automation are introduced, a comparative analysis of the tools at hand is performed before Cisco NDFC is introduced as the chosen orchestration platform for the purpose of the project. Lastly, virtualization/emulation is discussed, where definitions of both terms are introduced, the need for network emulation is justified as the chosen approach for testing the fabric in question, and why PNETLab is chosen as the testbed where the entire solution is deployed.

3.2 The Road to Network Automation

3.2.1 CLI Era and Early Monitoring

The development of network automation was not sudden; rather, it is the result of decades of continuous improvement. Network complexity, the need for speed in delivering services, and the adoption of SDNs are key factors that have led to network automation.
In the 1990s and early 2000s, most network device management activities were performed using CLI. This meant typing individual commands for individual devices. This approach is error-prone, non-scalable, and manual [41]. Monitoring activities were initially based on SNMP (Simple Network Management Protocol) and MIBs (Management Information Bases). These approaches provided read-only information on metrics such as interface counters and CPU loading averages but provided no native configuration management or programmatic control [42].

3.2.2 APIs and Programmability

This limitation of CLI and SNMP has prompted the advent of programmatic interfaces from the mid-2000s. NETCONF was introduced in RFC 6241, which provided a structured approach to configuration protocols using XML. Later on, RESTCONF was introduced as an alternative to the NETCONF protocol [43]. Scripting environments were introduced in network devices. For example, Cisco introduced the Embedded Event Manager, whereas Python was introduced in IOS XE. Around 2011, the advent of Open Flow and SDN has pushed the concept of the separation of control and data planes in the network. The network can be programmed using northbound APIs [15].

3.2.3 Configuration Management and IBN

As the API world evolved, the industry began to use existing configuration management tools such as Ansible, Puppet, and Chef, which were originally created to configure servers. These tools were applied to network devices. Ansible was noted for its agentless design and idempotent nature in the form of playbooks[44]. Intent-Based Networking (IBN) was introduced in the world of network automation in about 2017-2018. Instead of defining specific network operations, the network operator declares the network intent, such as "Ensure application X has 100 Mbps of connectivity between sites," and the network automatically figures out the required operations and ensures the network meets the intent[45]. The network state is constantly compared with the intended state.

3.2.4 The Modern Automation Stack

Nowadays, network automation is part of a complete technology stack consisting of:

- version control for configurations and automation scripts (Git).
- CI/CD pipelines for safe deployment of code changes (Jenkins, GitLab CI).
- Infrastructure as Code tools for managing network infrastructure (Terraform, Ansible, NSO).
- Observability, including telemetry and streaming analytics instead of traditional SNMP-based polling.
- AIOps, which uses machine learning for anomaly detection and root cause analysis [46].

This technology stack makes it possible to implement continuous delivery, fast rollback, and closed-loop remediation, turning networking into a proactive and software-driven discipline instead of a slow and manual craft.

Figure 3.1: Network Automation Timeline

3.3 Automation Architecture

3.3.1 Northbound and Southbound APIs

3.3.2 Orchestration vs Configuration Management

3.3.3 Declarative vs Imperative Models

3.4 Automation Tools

3.4.1 Selection Criteria

1. Ansible

2. Terraform
3. Cisco NSO
4. Cisco NDFC

3.4.2 Comparison Table

3.4.3 selected Solution Cisco NDFC

3.5 Cisco NDFC

3.5.1 History, Evolution and Positioning

3.5.2 Architecture and Key Capabilities

3.5.3 Lexicon

3.6 Virtualization and Emulation

3.6.1 Virtualization vs Emulation

3.6.2 Key Differences Between Virtualization and Emulation

3.6.3 Why Emulation for Network Engineering

3.6.4 Types of Network Emulators

3.7 Comparative Study of Emulation Platforms

3.7.1 Criteria

1. GNS3
2. EVE-NG
3. Cisco CML
4. PNETLab

3.7.2 Comparison Table

3.7.3 Selected Platform — EVE-NG

3.8 Conclusion

Chapter 4

Chapter 5

Netnography

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